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1. RECOLLECTIONS ON PERMUTATION GROUPS

Next we turn to a very important class of finite groups, the so-called permutation groups. You have probably seen these before, but I want to reintroduce the notation. Let

$$[n] := \{1, 2, \dots, n\}. \quad (1.1)$$

A permutation of $[n]$ is a bijection $p: [n] \rightarrow [n]$. It may be displayed in two-line notation as

$$p = \begin{pmatrix} 1 & 2 & \cdots & n \\ p(1) & p(2) & \cdots & p(n) \end{pmatrix}. \quad (1.2)$$

If p and q are permutations, we use the convention

$$qp = q \circ p; \quad (1.3)$$

thus the rightmost permutation is applied first. Let us also note that if p is a permutation, we write p^{-1} for the inverse permutation; that is, if $p(i) = k$, then $p^{-1}(k) = i$. One can check that

$$pp^{-1} = p^{-1}p = \text{id}, \quad (1.4)$$

where id is the identity permutation.

Let S_n denote the set of all permutations of $[n]$.

Proposition 1.1. *The triple $(S_n, \circ, \text{id})$ is a group.*

Proof. The composite of two bijections is again a bijection, so S_n is closed under composition. Composition of functions is associative, the identity map id is the identity element, and the inverse of a bijection is again a bijection. Hence all the group axioms hold. $\square$

2. CYCLE NOTATION

For future applications, we need to introduce the important ‘cycle notation’.

Definition 2.1. Fix a positive integer n and an integer k with $1 \leq k \leq n$. For distinct elements $a_1, \dots, a_k \in [n]$, the k -cycle $\sigma = (a_1 \cdots a_k)$ is the permutation defined by

$$\sigma(a_i) = a_{i+1} \quad (1 \leq i < k), \quad \sigma(a_k) = a_1. \quad (2.1)$$

Thus it sends $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_k \mapsto a_1$ and fixes every element outside $\{a_1, \dots, a_k\}$. Every one-cycle represents the identity permutation, which we denote by id or (1) . The empty product also represents the identity. A cycle of the form $(a \ b)$, with $a \neq b$, is called a *transposition*.

Remark. Cyclic rotations of the entries represent the same cycle. For example,

$$(1 \ 2 \ 3) = (2 \ 3 \ 1) = (3 \ 1 \ 2). \quad (2.2)$$

For example, the cycle

$$p = (1 \ 2 \ 3) \quad (2.3)$$

sends 1 to 2, 2 to 3, and 3 to 1, while fixing every other element. Thus, as a permutation of $[5]$,

$$(1 \ 2 \ 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}. \quad (2.4)$$

Example 2.1 (Composing cycles). Let $q = (2 \ 3 \ 4)$ and $p = (1 \ 5)$. Then

$$qp = (2 \ 3 \ 4)(1 \ 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 4 & 2 & 1 \end{pmatrix}. \quad (2.5)$$

When the entries in cycles are disjoint, the two permutations commute:

$$(2 \ 3 \ 4)(1 \ 5) = (1 \ 5)(2 \ 3 \ 4). \quad (2.6)$$

Cycles that are not disjoint need not commute. For example, you can verify that, as permutations of $[4]$, $(2 \ 4)(1 \ 4) \neq (1 \ 4)(2 \ 4)$. Indeed, we have

$$(2 \ 4)(1 \ 4) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}, \quad (1 \ 4)(2 \ 4) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix}. \quad (2.7)$$

Here is a basic fact:

Lemma 2.1. *Every permutation is a product of disjoint cycles. Also, every permutation is a product of (not necessarily disjoint) transpositions.*

Proof. We first prove that every permutation p of $[n]$ is a product of disjoint cycles. Choose $a_1 \in [n]$ and consider the sequence obtained by repeatedly applying p to a_1 , namely,

$$a_1, p(a_1), p^2(a_1), \dots \quad (2.8)$$

Since $[n]$ is finite, some term must eventually repeat. Let k be the first positive integer for which $p^k(a_1)$ is equal to an earlier term, say $p^k(a_1) = p^j(a_1)$ with $0 \leq j < k$. If $j > 0$, applying p^{-1} gives $p^{k-1}(a_1) = p^{j-1}(a_1)$, contradicting the choice of k . Hence $j = 0$.

It follows that

$$a_1, p(a_1), \dots, p^{k-1}(a_1) \quad (2.9)$$

are distinct and that p acts on them as the cycle

$$(a_1 \ p(a_1) \ \cdots \ p^{k-1}(a_1)). \quad (2.10)$$

If this cycle does not contain every element of $[n]$, choose an unused element a_2 and repeat the argument. The new cycle is disjoint from the first one: if the two cycles shared an element, repeatedly applying p or p^{-1} would show that a_2 lies in the first cycle, contrary to its choice. Repeating this process gives pairwise disjoint cycles containing every element of $[n]$. Their product agrees with p on every element, and therefore equals p . (Cycles of length one may be omitted.)

Moreover, each cycle of length $k \geq 2$ can be written as a product of transpositions:

$$(a_1 \ a_2 \ \cdots \ a_k) = (a_1 \ a_k)(a_1 \ a_{k-1}) \cdots (a_1 \ a_2). \quad (2.11)$$

To verify this identity, let T denote the product on the right and recall that the rightmost transposition is applied first. The rightmost transposition sends a_1 to a_2 , and all the remaining transpositions fix a_2 . Hence $T(a_1) = a_2$. Now suppose $2 \leq i < k$. The transpositions to the right of $(a_1 \ a_i)$ fix a_i . Then $(a_1 \ a_i)$ sends a_i to a_1 , and the next transposition, $(a_1 \ a_{i+1})$, sends a_1 to a_{i+1} . All the remaining transpositions fix a_{i+1} , so

$$T(a_i) = a_{i+1}. \quad (2.12)$$

Finally, all the factors except the leftmost one fix a_k , while the leftmost factor $(a_1 \ a_k)$ sends a_k to a_1 . Thus $T(a_k) = a_1$. Every element outside $\{a_1, \dots, a_k\}$ is fixed by every factor of T . Therefore T has exactly the same action as $(a_1 \ a_2 \ \cdots \ a_k)$, proving the identity.

It follows that every permutation is a product of transpositions. The identity permutation is represented by the empty product.

□

Example 2.2 (Disjoint cycles in S_5). Consider

$$p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 5 & 2 & 3 \end{pmatrix}. \quad (2.13)$$

Starting with 1 and following its images gives

$$1 \mapsto 4 \mapsto 2 \mapsto 1, \quad (2.14)$$

while the remaining elements that are not yet accounted for satisfy

$$3 \mapsto 5 \mapsto 3. \quad (2.15)$$

Thus the decomposition of p into disjoint cycles is

$$p = (1 \ 4 \ 2)(3 \ 5). \quad (2.16)$$

Example 2.3 (Transposition decompositions are not unique). Consider

$$q = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 1 & 5 & 3 \end{pmatrix}. \quad (2.17)$$

Consider the cycle

$$q = (1 \ 2 \ 4 \ 5 \ 3). \quad (2.18)$$

The factorization used in the proof of the lemma gives

$$q = (1 \ 3)(1 \ 5)(1 \ 4)(1 \ 2). \quad (2.19)$$

However, the same cycle also has the different factorization

$$q = (1 \ 2)(2 \ 4)(4 \ 5)(5 \ 3). \quad (2.20)$$

One can check either product from right to left: both send $1 \mapsto 2$, $2 \mapsto 4$, $4 \mapsto 5$, $5 \mapsto 3$, and $3 \mapsto 1$. Thus a permutation's decomposition into transpositions is not unique.

3. THE GROUP S_3

The six elements of S_3 are the identity, the three transpositions, and the two 3-cycles:

$$S_3 = \{\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}. \quad (3.1)$$

Set

$$x = (1\ 2\ 3) \quad \text{and} \quad y = (1\ 3). \quad (3.2)$$

Then

$$x^2 = (1\ 3\ 2), \quad x^3 = \text{id}, \quad y^2 = \text{id}, \quad (3.3)$$

and direct calculation gives

$$yx = x^2y. \quad (3.4)$$

Consequently,

$$S_3 = \{\text{id}, x, x^2, y, xy, yx\}. \quad (3.5)$$

Sometimes one writes down a group by specifying its entire multiplication table. For S_3 this is shown below. The entry in row g and column h is the product gh , so the permutation in column h is applied first.

$\cdot$	id	x	x^2	y	xy	yx
id	id	x	x^2	y	xy	yx
x	x	x^2	id	xy	yx	y
x^2	x^2	id	x	yx	y	xy
y	y	yx	xy	id	x^2	x
xy	xy	y	yx	x	id	x^2
yx	yx	xy	y	x^2	x	id

(3.6)

Since

$$x = (1\ 2\ 3), \quad x^2 = (1\ 3\ 2), \quad y = (1\ 3), \quad xy = (2\ 3), \quad yx = (1\ 2), \quad (3.7)$$

the same multiplication table in cycle notation is

$\cdot$	id	(1 2 3)	(1 3 2)	(1 3)	(2 3)	(1 2)
id	id	(1 2 3)	(1 3 2)	(1 3)	(2 3)	(1 2)
(1 2 3)	(1 2 3)	(1 3 2)	id	(2 3)	(1 2)	(1 3)
(1 3 2)	(1 3 2)	id	(1 2 3)	(1 2)	(1 3)	(2 3)
(1 3)	(1 3)	(1 2)	(2 3)	id	(1 3 2)	(1 2 3)
(2 3)	(2 3)	(1 3)	(1 2)	(1 2 3)	id	(1 3 2)
(1 2)	(1 2)	(2 3)	(1 3)	(1 3 2)	(1 2 3)	id

(3.8)

In particular, S_3 is not abelian: for example, $xy \neq yx$.

Example 3.1 (A permutation of eight letters). Consider

$$\sigma = (1\ 5\ 7)(2\ 6)(3\ 4) \in S_8. \quad (3.9)$$

The cycles are disjoint, and therefore

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 4 & 3 & 7 & 2 & 1 & 8 \end{pmatrix}. \quad (3.10)$$

Thus $1 \mapsto 5$, $5 \mapsto 7$, $7 \mapsto 1$; the elements 2 and 6 are interchanged, as are 3 and 4; and 8 is fixed.

Cayley graph of S_3 .

A Cayley graph gives a geometric picture of a group together with a chosen set of ‘generators.’ We use

$$x = (1\ 2\ 3) \quad \text{and} \quad y = (1\ 3). \quad (3.11)$$

The vertices are the six elements of S_3 . From each vertex g , a blue arrow leads to gx , and a red edge joins g to gy . The red edges need no arrows because $y^2 = \text{id}$: traversing a red edge twice returns to the starting vertex. The blue edges form two directed triangles, and the red edges join corresponding vertices of the two triangles.

